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Stability and functional evolution of a novel family of hyperstable apple snail egg proteins

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ABSTRACT

The relationship between protein stability and function evolution in nature has not been explored in detail. We investigate this fundamental question by characterizing the Perivitellin-1 (PV1) orthologs within *Pomacea* snails. These highly stable egg carotenoproteins, crucial to the snail's reproductive success, have acquired clade-related protective functions. To gain insight into the molecular evolution of reproductive proteins along a genus we studied *P. patula* PV1 (PpaPV1) from Flagellata eggs, the most basal clade of *Pomacea*. PpaPV1 stands as the most stable PV1, with a much longer unfolding half-life and resistance to detergent unfolding, indicating a loss of kinetic stability in the derived clades. In fact, PpaPV1 is among the most hyperstable proteins described in nature. Its molecular weight, size, and high resistance to proteolysis are similar to and shared with all PV1s. Its spectral characteristics and associated pale egg coloration as well as a mild lectin activity indicate these protein traits appeared early in the evolution of the genus. However, lectin binding is less strong, and glycan specificity is narrower than those PV1s of its sister Bridgesii clade.

Our results provide experimental evidence of the relationship between protein stability and evolvability, showing that along their rapid evolution, PV1s tradeoff between structural and

kinetic stability loss and the acquisition of diverse defensive traits: a strong lectin activity (Bridgesii clade), or an aposematic warning coloration (Canaliculata clade). This study offers insight into the molecular evolution of a novel family of invertebrate reproductive proteins that accompanied the fast diversification of the genus and may have contributed to Canaliculata snails invasiveness.

Competing Interest Statement

The authors have declared no competing interest.

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